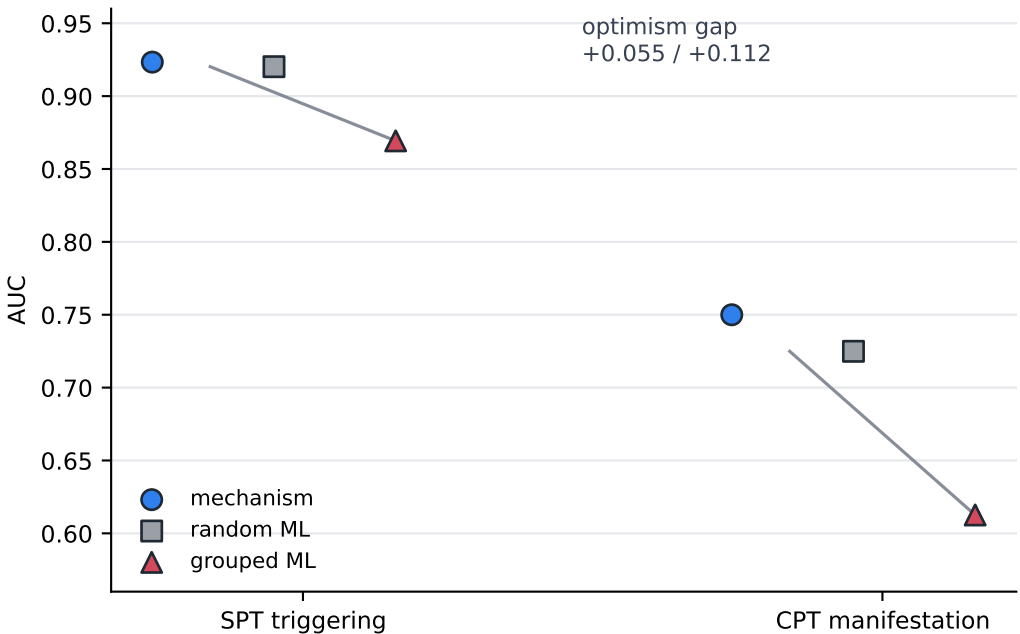
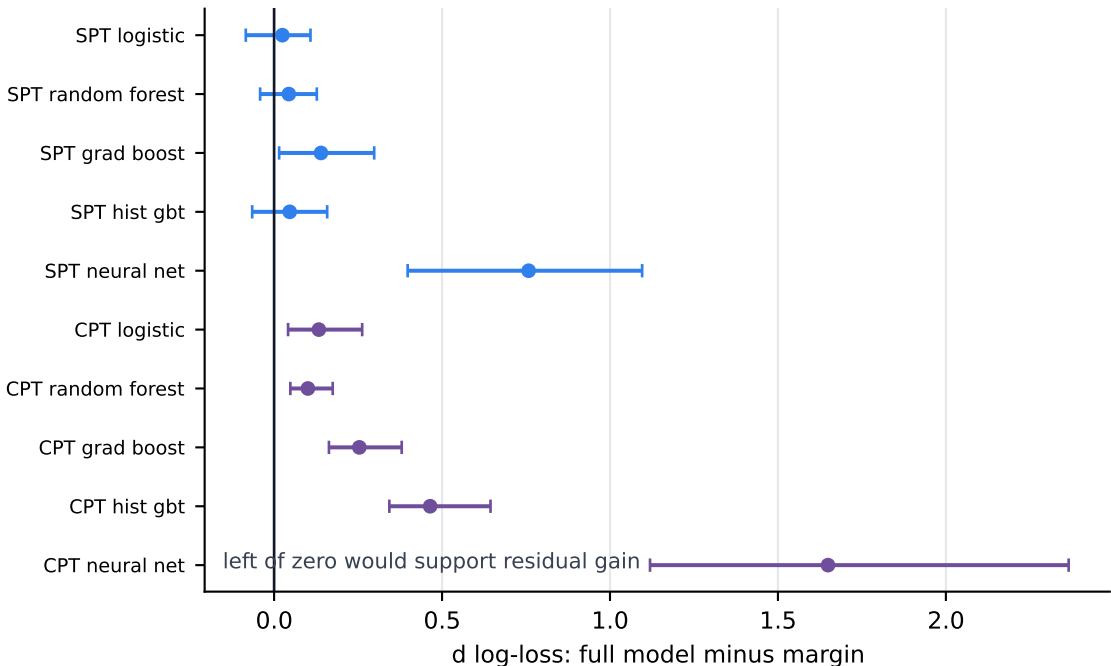


Operational non-inferiority under earthquake-grouped validation

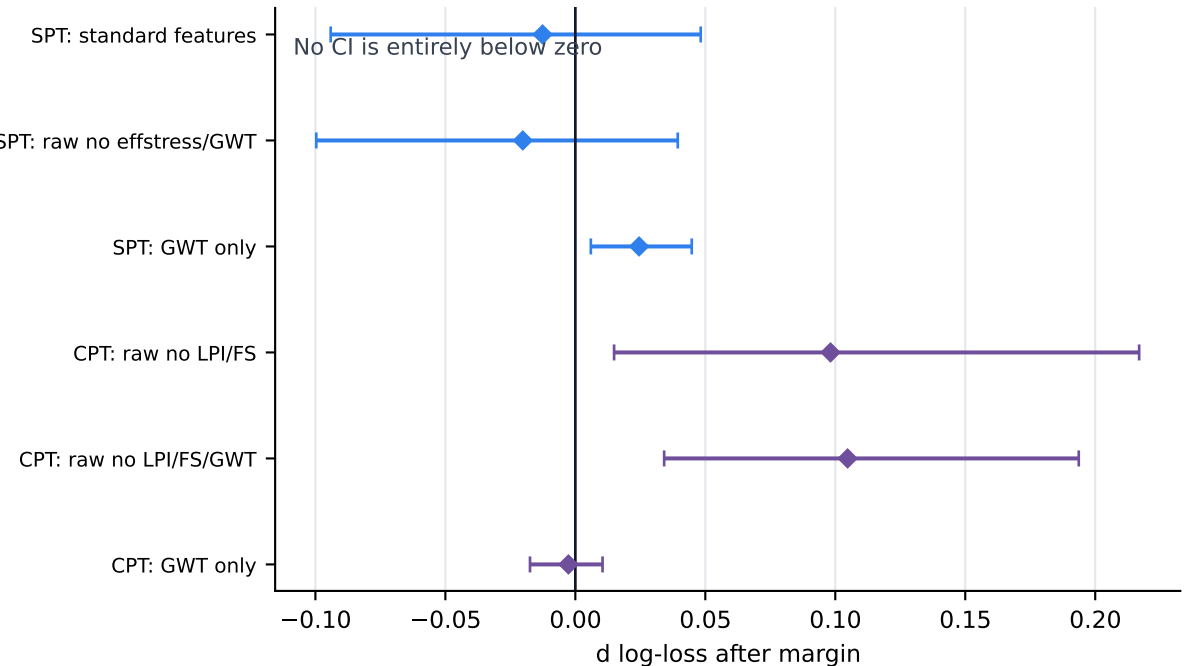
A Random validation inflates apparent ML skill



B Full models do not improve OOS likelihood



C Residual/raw variables after the margin



D Practical-improvement audit bounds the claim

